

Bias Ablation -- Synthesis Report

4DVarNet-FM: H1 Bias Ablation Report

Sweeps param_bias and forcing_state_bias independently on a case=2 (corrupted, quartic-coupled forcing) dataset and evaluates the 4 baseline DA methods. CS1-CS4 are untouched by this ablation.

Common parameters:

T_max = 3.0s dt = 0.01 obs_interval = 20
R_var = 0.5 B_var = 2.0 forcing_coupling = quartic

DA Baseline parameters:

DA window steps (DWS) = 50 batch_size = 128
Weak-4DVar : opt_steps=150, lr=0.02
Strong-4DVar: max_iter=40, lr=0.1
EnKF : inflation=1.2
ETKF : inflation=1.2

Test data:

200 trajectories per case, generated with seed=200+i (i = case index)
Forcing corrupted by Ornstein-Uhlenbeck noise (tau_eta=5.0, sigma_eta=0.707), quartic coupling

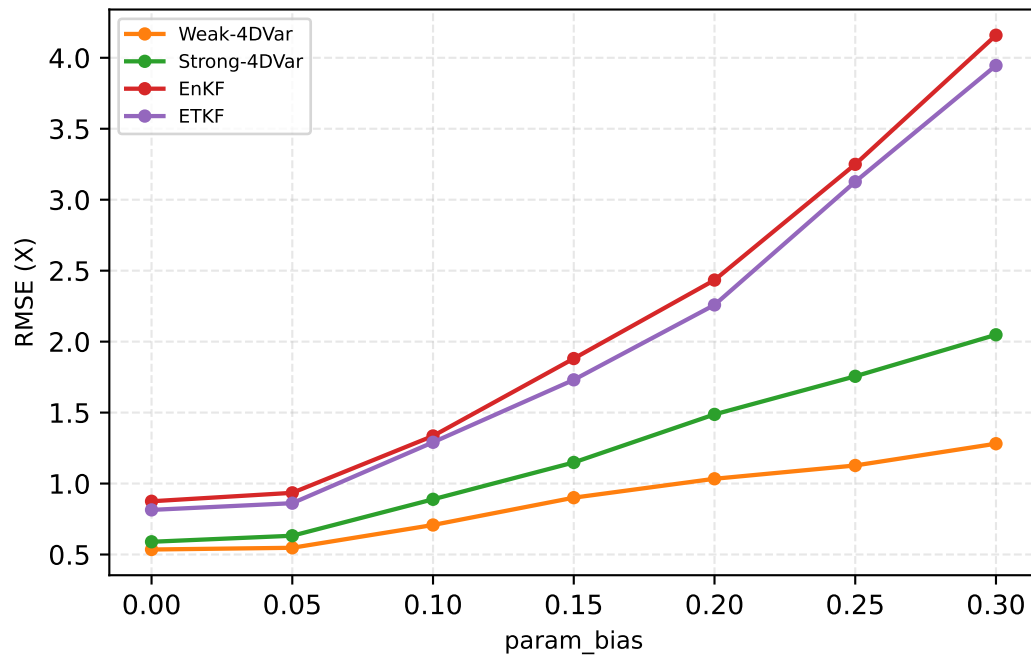
Bias Ablation -- Mean RMSE per Case

param_bias	forcing_state_bias	Weak-4DVar	Strong-4DVar	EnKF	ETKF
0.00	0.00	0.6734	0.7503	1.1728	1.0715
0.05	0.00	0.8022	0.9729	1.4091	1.3146
0.10	0.00	1.2073	1.5351	2.1665	2.0854
0.15	0.00	1.6570	2.1355	3.0609	2.8823
0.20	0.00	2.1139	2.7829	3.9803	3.7739
0.25	0.00	2.4869	3.3594	5.2159	5.0681
0.30	0.00	2.9258	3.9672	6.3893	6.1803
0.00	0.05	0.6660	0.8411	1.2747	1.1618
0.00	0.10	0.8113	0.9753	1.4067	1.2819
0.00	0.15	0.6847	0.8276	1.5478	1.3734
0.00	0.20	0.8490	0.9296	1.7750	1.5426
0.00	0.25	0.8370	1.0427	2.0052	1.7828
0.00	0.30	0.9497	1.1422	2.6327	2.2882

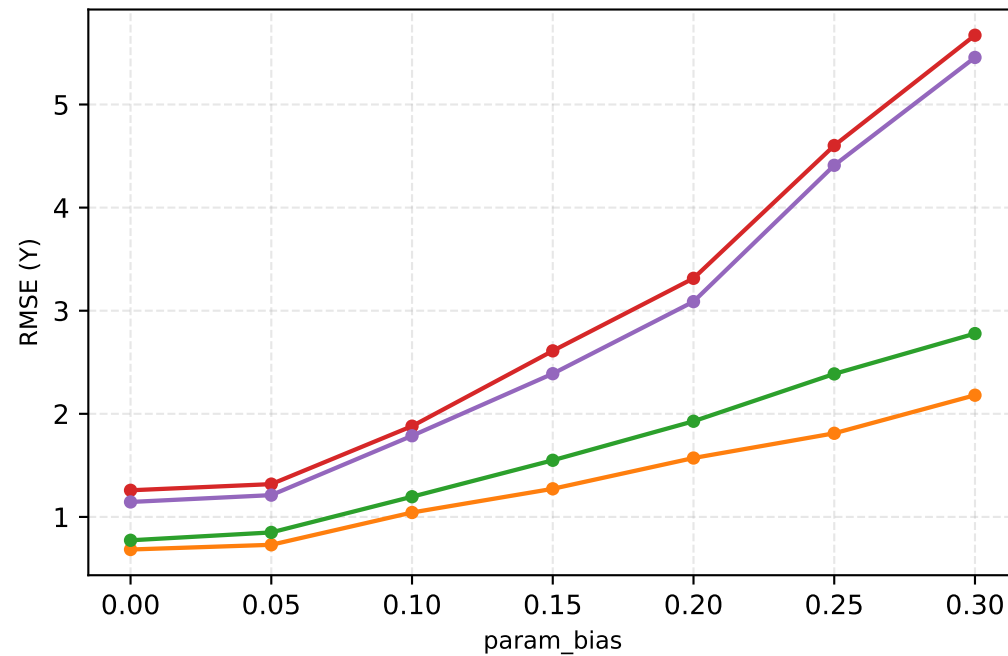
Config file: experiments/H1_bias_ablation/results.json

RMSE vs. param_bias (forcing_state_bias = 0)

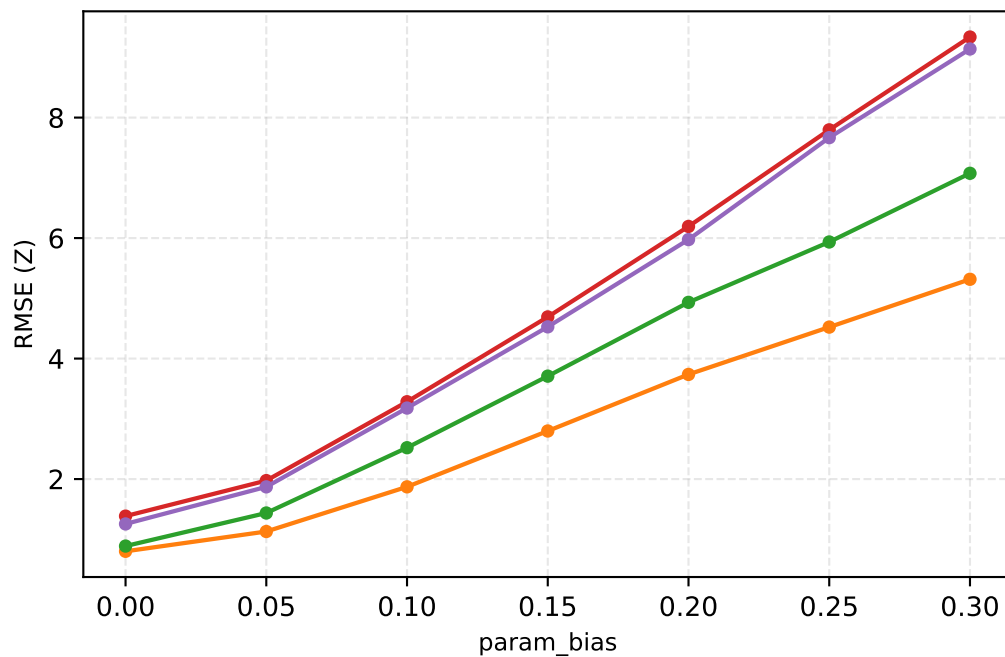
X



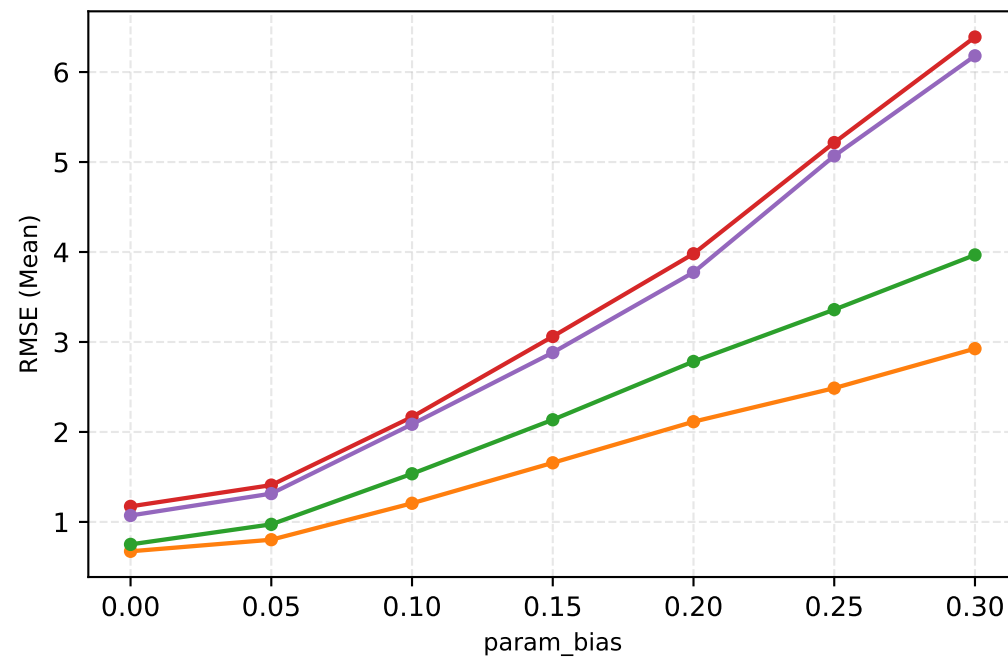
Y



Z



Mean



RMSE vs. forcing_state_bias (param_bias = 0)

